Streamflow

Katie Markovich, EPS 522/ ENVS 423L (Fall 2025)

Housekeeping

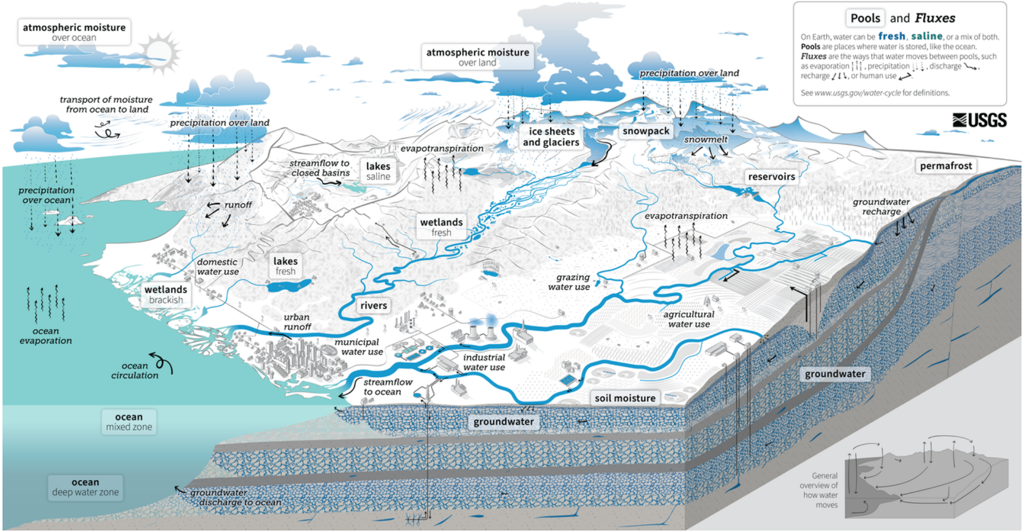
- Problem sets should be coming back to you soon
- PS#4 due Thursday by midnight
- PS #4 problem 1.3 should be "units of meters" (volume should obviously not be square meters)
- PS #4 problem 1.4: pick one calendar year and download the data (csv) for each of the models to get the variation in cumulative ET values. Uncertainty is standard deviation of those estimates.
- Thursday is a guest lecture on ET
- No lab Thursday!

522 Term Project Lit. Review

1. Preliminary water balance equation
2. Review academic/grey literature for studies relating to any component of the water balance in the region of your control volume (or a related region):
    - What did they do?
    - What did they find?
    - What were the limitations?
3. How will this previous work inform your hydrologic analysis and what gaps remain that you aim to address?
    - Could use their methods
    - Could use their water balance (component) as a starting point or point of validation

Not looking for multiple pages here, a page or two is fine!

Water Cycle



atmospheric moisture
over ocean
transport of moisture
from ocean to land
precipitation over land
atmospheric moisture
over land
precipitation over land
ice sheets
and glaciers
snowpack
snowmelt
evapotranspiration
streamflow to
closed basins
lakes
saline
permafrost
reservoirs
precipitation
over ocean
runoff
wetlands
fresh
evapotranspiration
groundwater
recharge
domestic
water use
lakes
fresh
wetlands
brackish
grazing
water use
rivers
agricultural
water use
ocean
evaporation
urban
runoff
municipal
water use
industrial
water use
ocean
circulation
streamflow
to ocean
groundwater
soil moisture
groundwater
ocean
mixed zone
ocean
deep water zone
groundwater
discharge to ocean
General
overview of
how water
moves
Pools and Fluxes
On Earth, water can be fresh, saline, or a mix of both.
Pools are places where water is stored, like the ocean.
Fluxes are the ways that water moves between pools, such as evaporation, precipitation, discharge, recharge, or human use.
See www.usgs.gov/water-cycle for definitions.
USGS

Learning Objectives

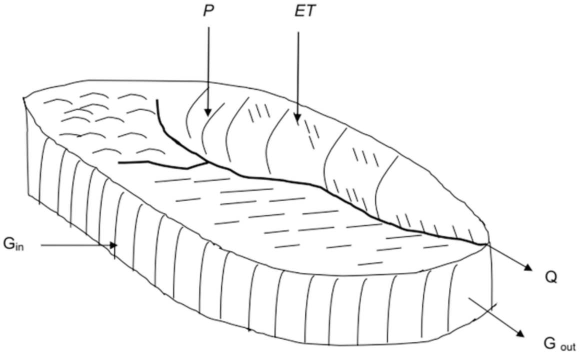
1. Stream response to water input
2. Basics of stream gaging
3. Hydrograph separation purpose and methods
4. Runoff-generation mechanisms
5. Stream network and routing
6. Rainfall runoff-modeling

Regional Water Balance

$$\hat{P} + G_{in} - (Q + ET^{*} + G_{out}) = \Delta S$$

where ET^{*} is:

- free-water evap
- bare-soil evap
- transpiration
- interception loss



Where:

P = Precipitation
ET = Evapotranspiration
Q = Stream outflow
G_{in} = Ground water in
G_{out} = Ground water out

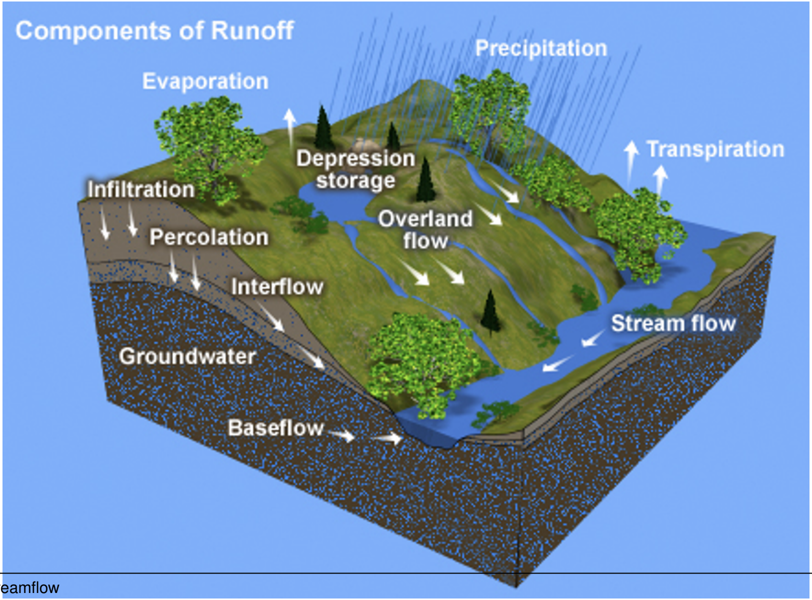
Importance of Stream Response

Streams are the routes for P minus ET (precipitation excess) on the continents that is either returned to the oceans or, in closed basins, evaporated.

Understanding stream response to water-input events is critical for:

- Water supply
- Flood prediction and forecasting
- Water quality

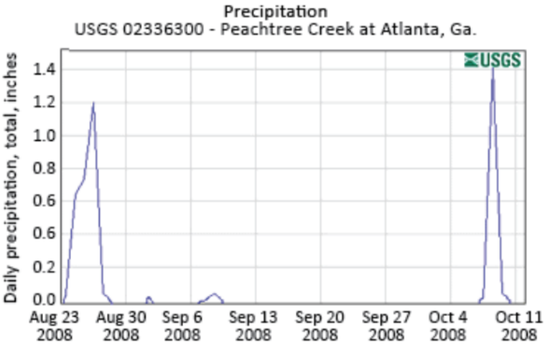
Flowpaths

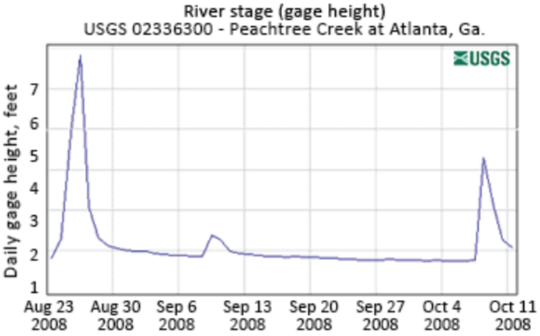


Components of Runoff
Precipitation
Evaporation
Transpiration
Depression storage
Infiltration
Overland flow
Percolation
Interflow
Stream flow
Groundwater
Baseflow

Baseflow

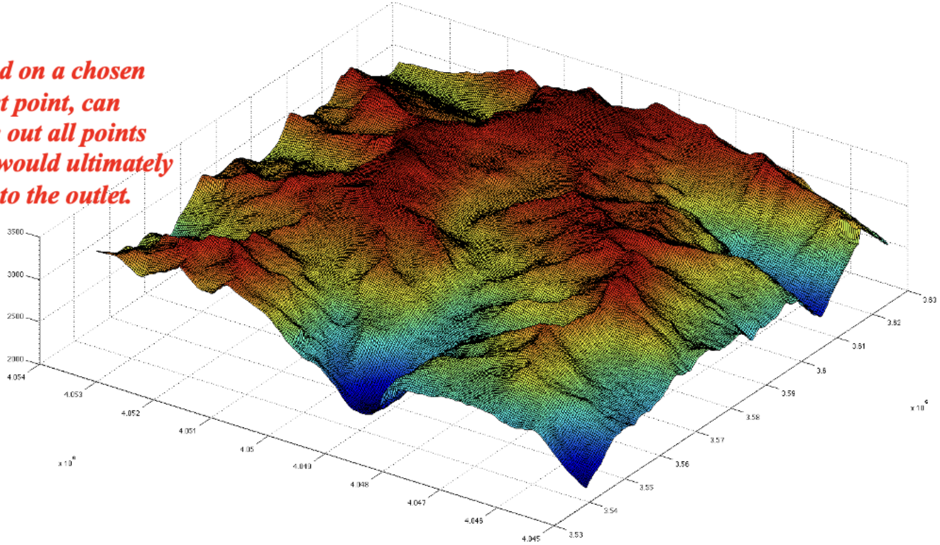
Baseflow is groundwater flow to streams. It is why streams have flow in them long after any rain event in the drainage area.





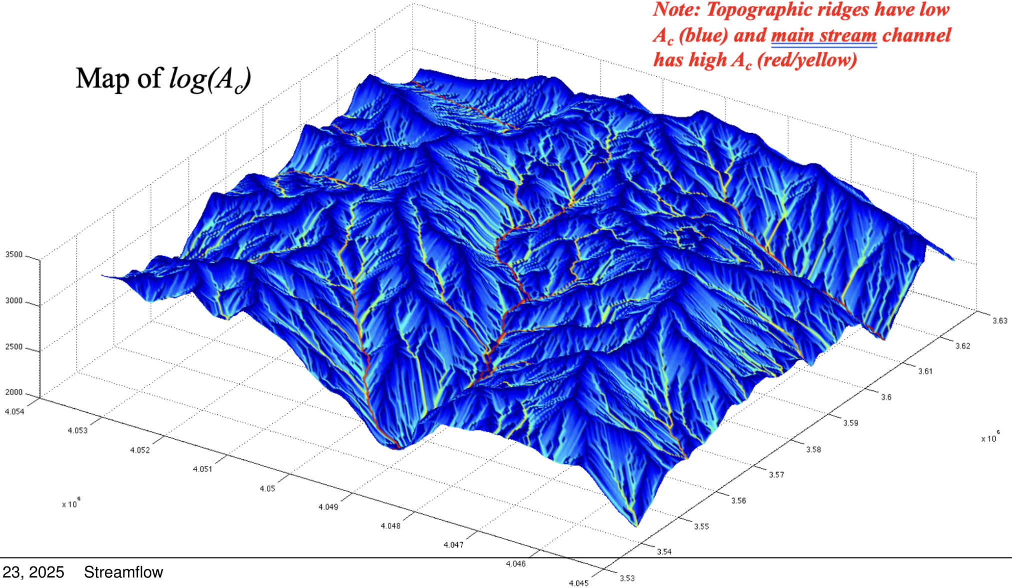
Upstream Contributing Area

Based on a chosen outlet point, can trace out all points that would ultimately flow to the outlet.

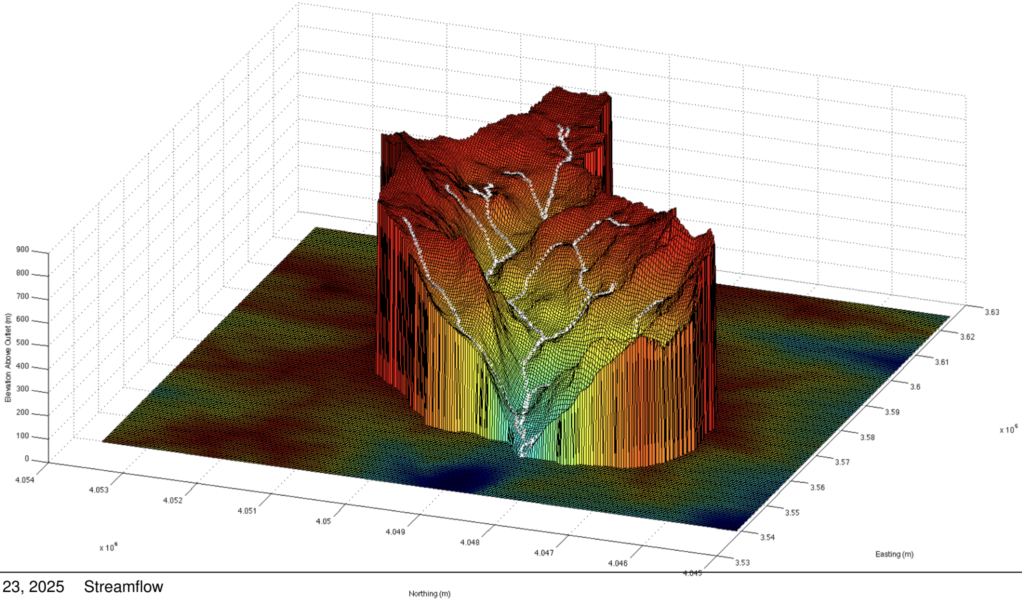


Upstream Contributing Area

Note: Topographic ridges have low A_c (blue) and main stream channel has high A_c (red/yellow)

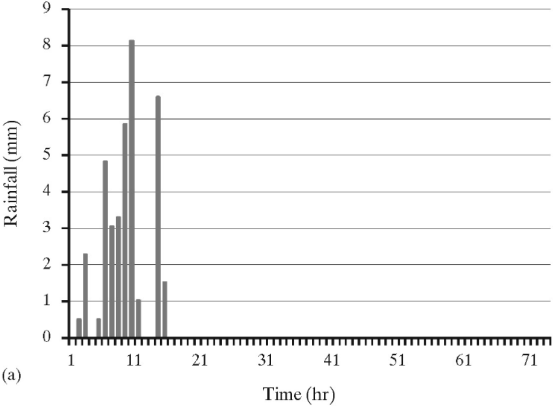


Upstream Contributing Area



Stream Response Basics

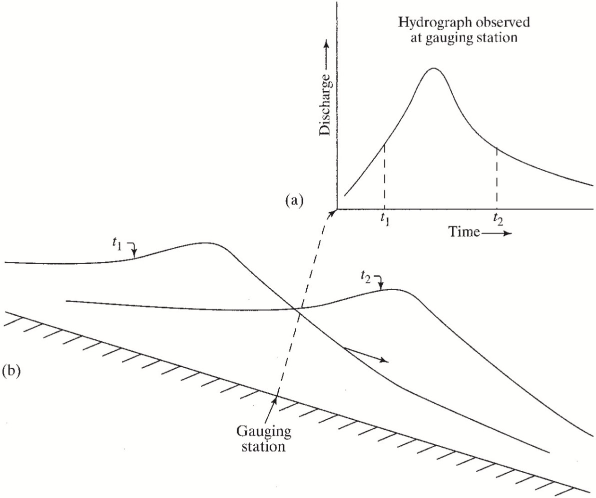
Stream response requires characterization of the water input (i.e., precipitation) which can be spatially averaged to produce a **hyetograph** in units of length vs. time.



(a)

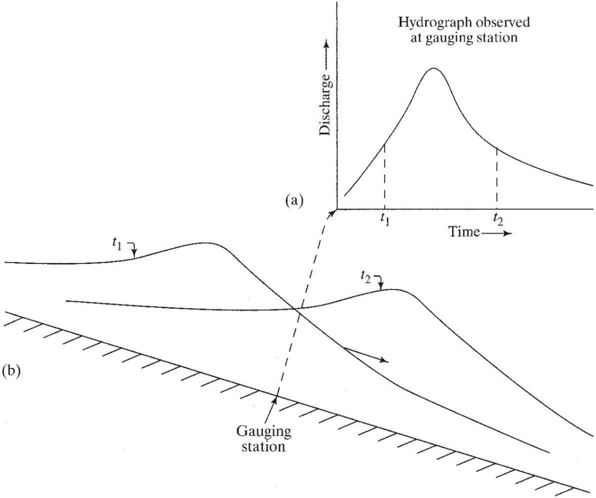
Stream Response Basics

Watershed response to the input event is shown via a **hydrograph**, or graph of discharge vs. time for a fixed point along the stream channel. Can be converted to L/T to compare directly to hyetograph.



Stream Response Basics

A hydrograph responding to an isolate input event shows a characteristic **rise to peak flow followed by a recession** back to its pre-event value.



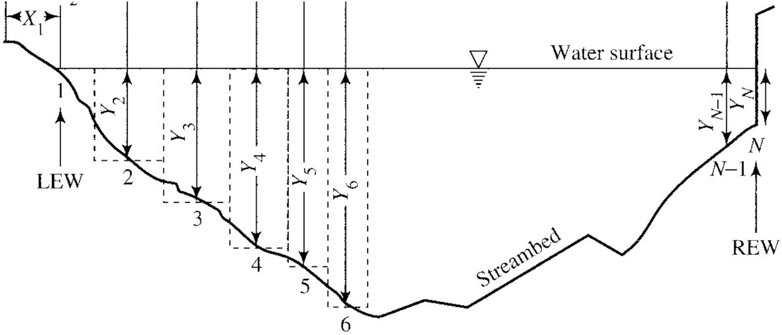
Stream Gaging

1. Direct measurement
   - Velocity-Area
   - Dilution
2. Indirect measurement
   - Empirical rating curve (natural control)
   - Theoretical rating curve (artificial control)

Velocity Area Method

$$Q = U \cdot A$$

where Q is discharge (L^3/T), U is velocity (L/T) and A is cross-sectional area. Direct measurement of stream velocity (using a flowmeter) along a stream cross-section and integration of values to give total discharge.




Alt Text: A person holding a yellow toy and a child...

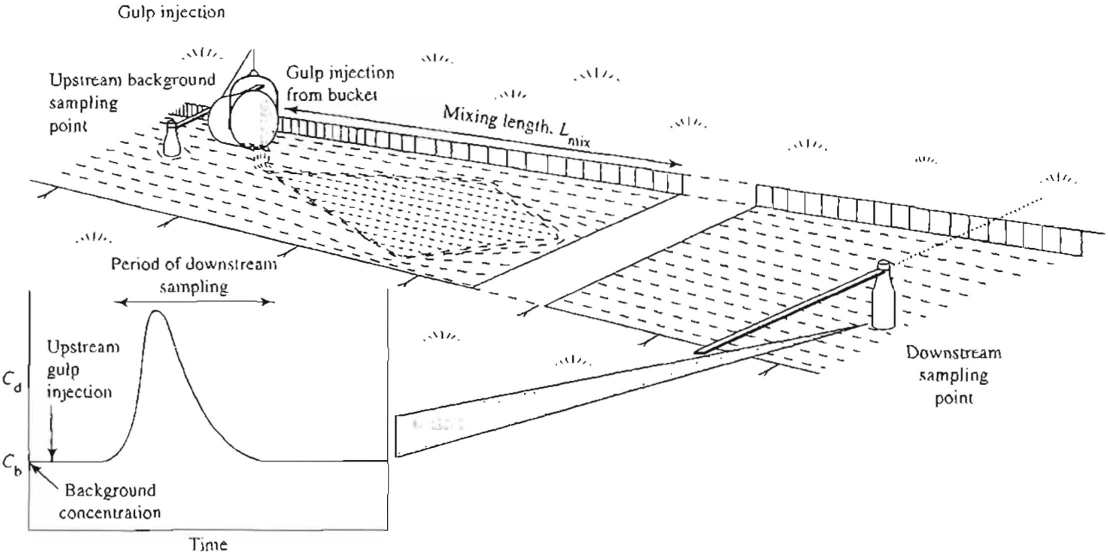
Dilution Method

Best for small, highly turbulent streams with rough irregular channels. Assumption that discharge is constant during test and between injection and observation point.

Most common to add slug of conservation tracer (salt) of known volume (V_t) and concentration (C_t) and monitor downstream site concentrations (C_d(t)) until they recede to background concentration (C_b).

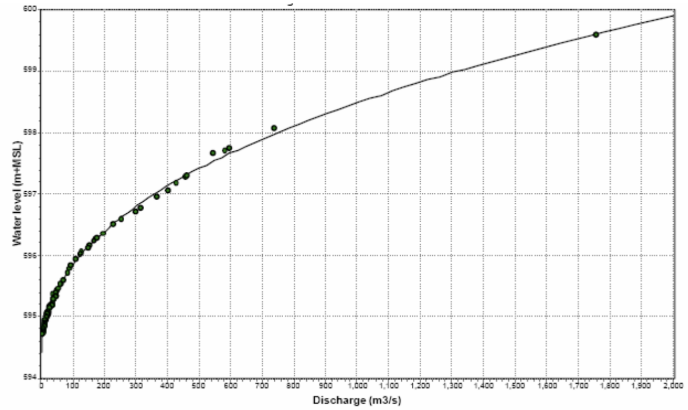
$$Q = \frac{(C_t - C_b) \cdot V_t}{\int_0^\infty [C_d(t) - C_b] \cdot dt}$$

Dilution Method



Empirical Rating Curve

Direct measurement is difficult and time-consuming, and impossible to record over time. Water surface elevation, or **stage**, is relatively simple to record using a pressure transducer. Stage can be related to discharge via an empirical rating curve:



Theoretical Rating Curve

We can theoretically determine discharge via artificial controls (**weirs**).

V-notch Weir

For a fully contracted, 90 degree, sharp crested V-notch weir, discharge (in cfs) can be determined from height (H, in ft) as:

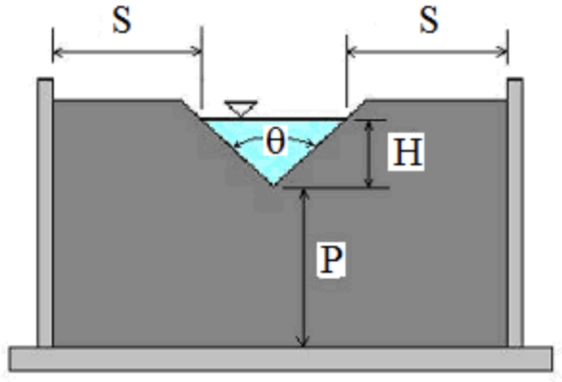
$$Q = 2.49 \cdot H^{2.48}$$

subject to:

$$P \geq 2 \cdot H_{max}$$

$$S \geq 2 \cdot H_{max}$$

$$0.2ft \leq H \leq 1.25ft$$



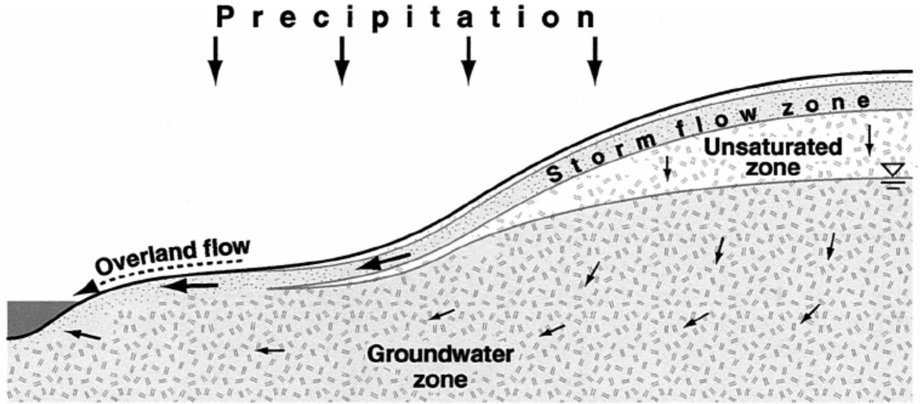
Stream Response Basics

Streamflow response is a spatially and temporally integrated response that is scale-dependent and determined by:

1. Spatially and temporally varying input rates
2. Travel time of a drop of water to reach the stream network (determined by length, slope, vegetation, soil, and geology)
3. Travel time for water to travel from entrace to the point of measurement (determined by length and nature of channel network)

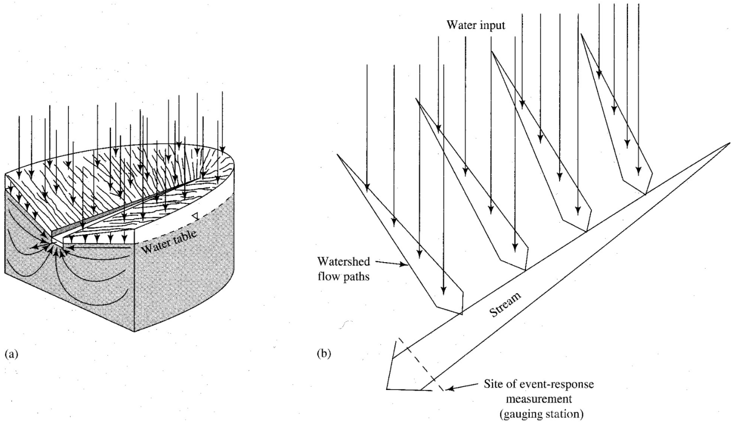
Hillslope Scale

Flow is the accumulation of **lateral** in flows along infinite number of surface and subsurface flowpaths of varying length and character.



Stream Network Scale

Accumulation of tributary flow which each reflect flow accumulation from hillslopes of varying length and character. Flow in the stream takes the form of a **flood wave** moving through the stream network.



National Water Model example

Complexities of Stream Response

- Volume of water appearing in the response hydrograph is typically a small fraction of volume that fell or melted (due to ET and recharge to slower groundwater flowpaths).

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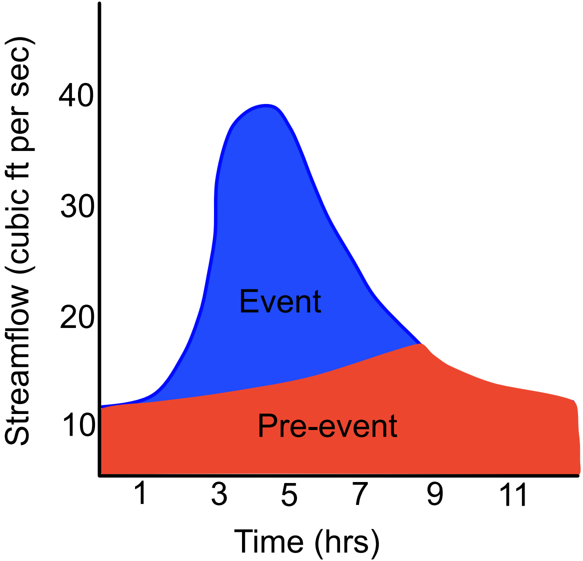
- Volume of water appearing in the response hydrograph is typically a small fraction of volume that fell or melted (due to ET and recharge to slower groundwater flowpaths).
- Volume of water appearing in the response hydrograph is not necessarily the same water that fell or melted as part of the input event.
- This motivates interest in separating the hydrograph into various components of flow, the simplest being **event**, or water that entered as part of the input event, and **pre-event flow**, or water that existed in the watershed prior to the water input event.

Graphical Hydrograph Separation



$$Q_{stream} = Q_{event} + Q_{pre-event}$$

Draw a line from the inflection point and measure the area to estimate Q_{event}

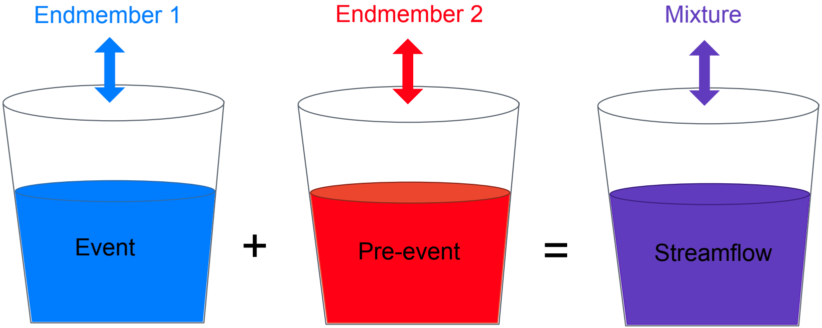
Issues?

Digital Filter Methods

Separate hydrograph based on frequency of sources, where event flow is high and pre-event is low frequency. More sophisticated than graphical method, but still heavily rely on simplifying assumptions.

`https://sephydro.hydrotools.tech/pageMain.php`

Endmember Mixing Analysis



$$(Q_{event} \times C_{event}) + (Q_{pre\text{-}event} \times C_{pre\text{-}event}) = (Q_{stream} \times C_{stream})$$

Q= streamflow (volume/time)

C= concentration (mass/volume)

Endmember Mixing Analysis

Two equations with two unknowns can be solved as follows:

$$Q_{stream} = Q_{pre-event} + Q_{event}$$

$$Q_{stream}C_{stream} = Q_{pre-event}C_{pre-event} + Q_{event}C_{event}$$

Endmember Mixing Analysis

Two equations with two unknowns can be solved as follows:

$$Q_{event} = \frac{C_{stream} - C_{pre-event}}{C_{event} - C_{pre-event}} \cdot Q_{stream}$$

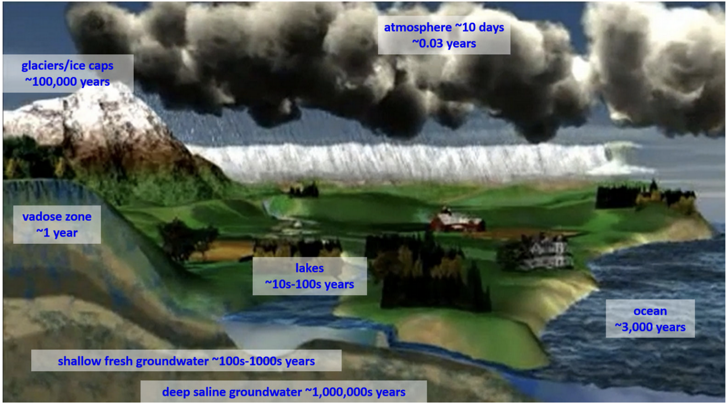
$$Q_{pre-event} = Q_{stream} - Q_{event}$$

Assumptions:

1. There is a significant difference between the concentration of different endmembers.
2. The concentration of endmembers is constant in space and time, or variations are accounted for.
3. Contributions from surface storage are negligible.

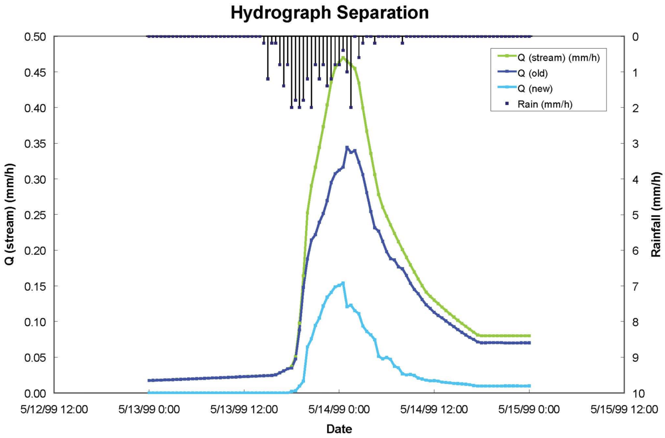
Residence Time Refresher

Stores are never static, and the rate at which they "turn over" is referred to as **residence time.** More specifically, residence time is the average length of time a parcel of water spends in the store.



Old Water Paradox

Pre-event water has longer residence times than event water, and is thus referred to as "old water." The observation that a storm hydrograph in response to an input event of new water is dominated by old water is referred to as the "Old Water Paradox."



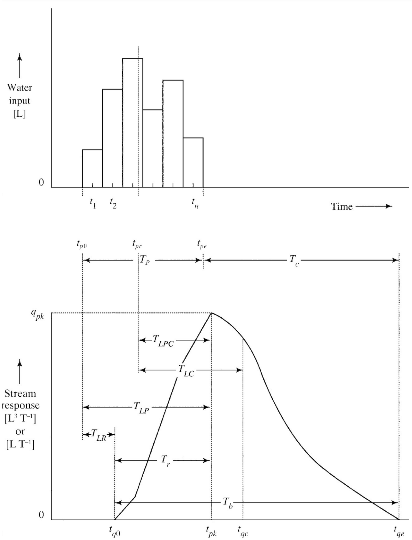
Streamflow Response Terms

Hyetograph Terms:

- Effective Water Input, W_{eff} = W - "losses"
- Center of mass time, t_{wc}, or time at which 50% of the input has occurred

Hydrograph Terms:

- Peak flow, q_{pk}
- Peak time, t_{pk}
- Response lag, $T_{LR} = t_{q0} - t_{w0}$
- Lag-to-peak, $T_{LP} = t_{pk} - t_{w0}$
- Center of mass time, t_{qc}
- **Centroid lag**, $T_{LC} = t_{qc} - t_{wc}$



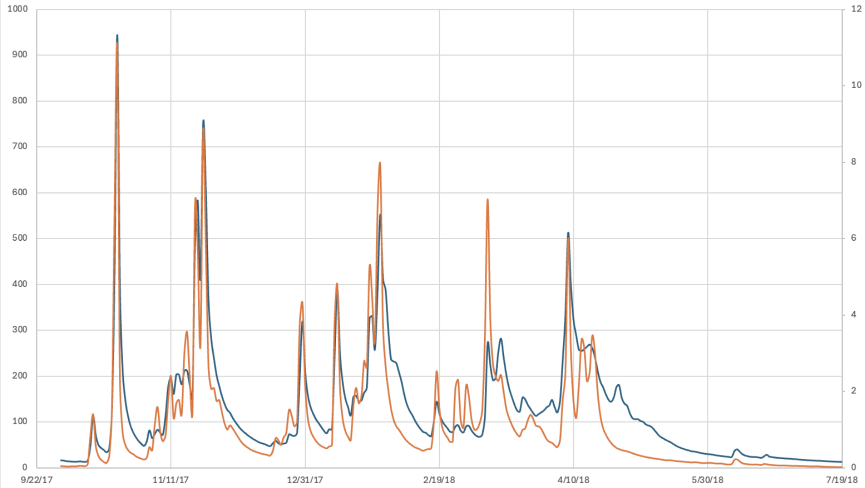
Factors Affecting Centroid Lag

Centroid lag is a useful metric for characterizing the overall response time of a watershed.

- Watershed size:
- Soils and Geology:
- Slope:
- Land use:

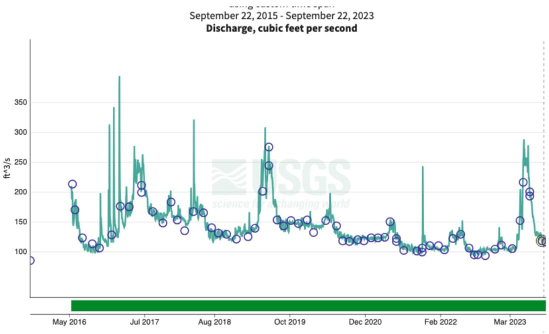
Guess that hydrograph!

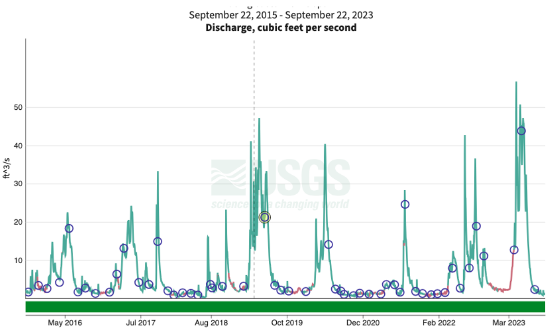
Which watershed is bigger?



Guess that hydrograph!

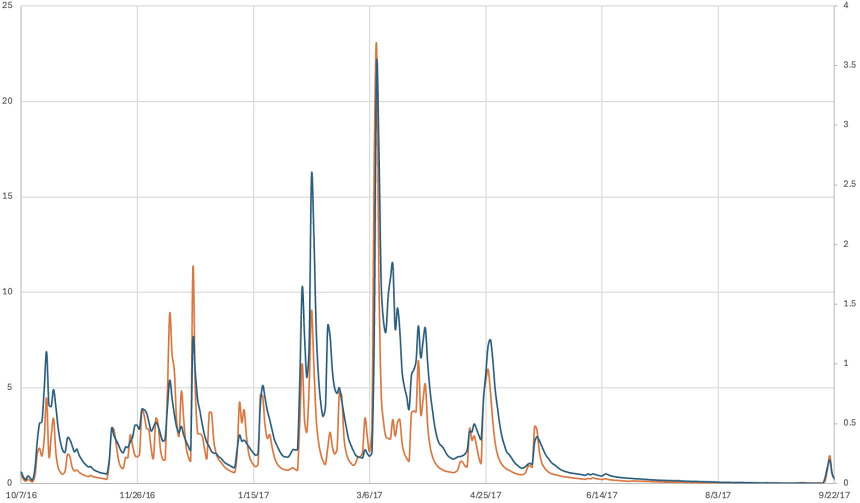
Which one has more permeable soils/bedrock?





Guess that hydrograph!

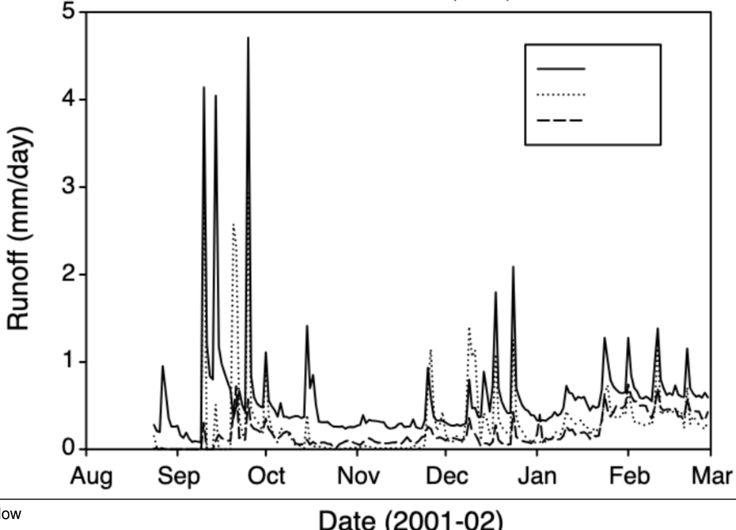
Which one is steeper?



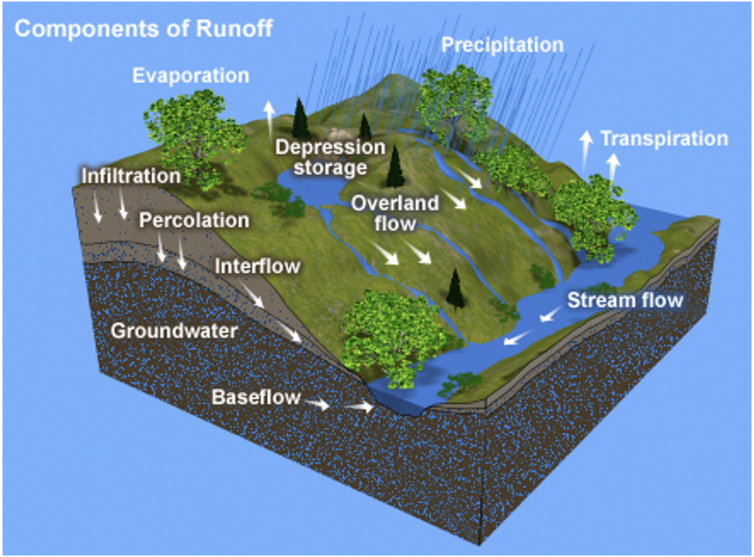
Guess that hydrograph!

Low, medium, high degree of urbanization?

from Burns et al. (2005) *JoH*



Runoff-Generation Mechanisms



1. Channel precipitation
2. Overland flow
3. Subsurface event flow

Channel Precipitation

Precipitation that falls directly on stream channel.

$$W_{cp} = W \cdot A_{cn}$$

where W is the average water input rate across the drainage area and A_{cn} is the area of

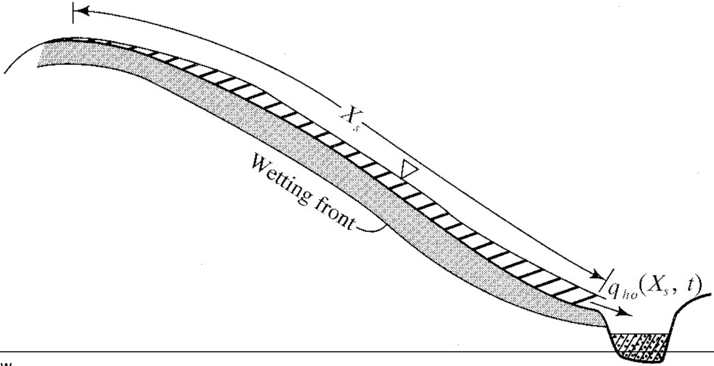
the channel network, typically a small fraction (1%) of the total drainage area. Despite the small area, it can be a significant component of q_{pk} and event flow.

Hortonian Overland Flow

"Saturated from above" or "infiltration-excess" overland flow.

$$q_{ho}(t) = W_{eff}(t) - f(t)$$

where $f(t)$ is the infiltration rate, governed by the saturated hydraulic conductivity (K_h^*) of the surface layer.

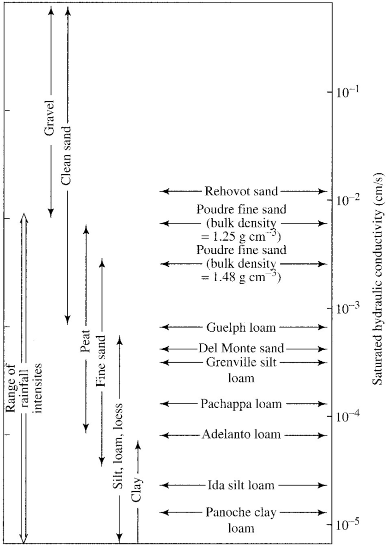


Hortonian Overland Flow

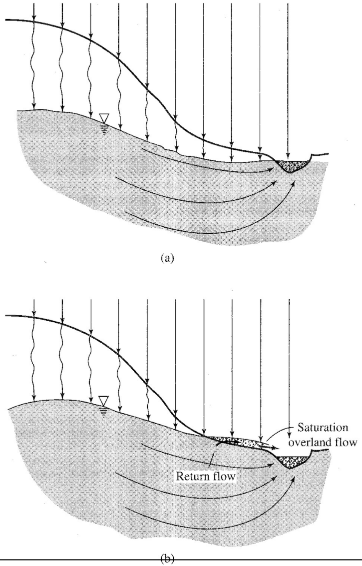
Hortonian overland flow was thought to be the principal runoff-generating mechanism from the 1930s through the 1950s, but it turns out infiltration-excess runoff is quite rare.

Wait, what is hydraulic conductivity?

Hydraulic conductivity (K), having units of L/T, is the ability of a porous medium to transmit water. Saturated hydraulic conductivity is used for unsaturated porous media, like soils, which can have varying degrees of saturation.



Saturation Overland Flow

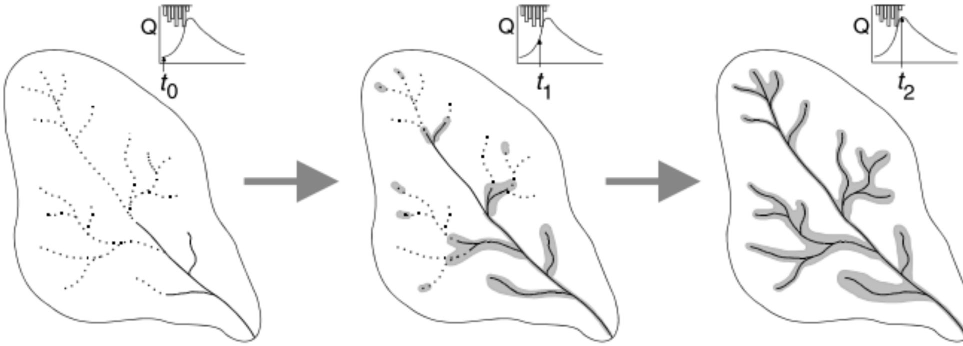


Also called Dunnian overland flow, where water input infiltrates and saturated zone "breaks out" of the land surface.

First observed in humid regions with shallow water table and gaining streams.

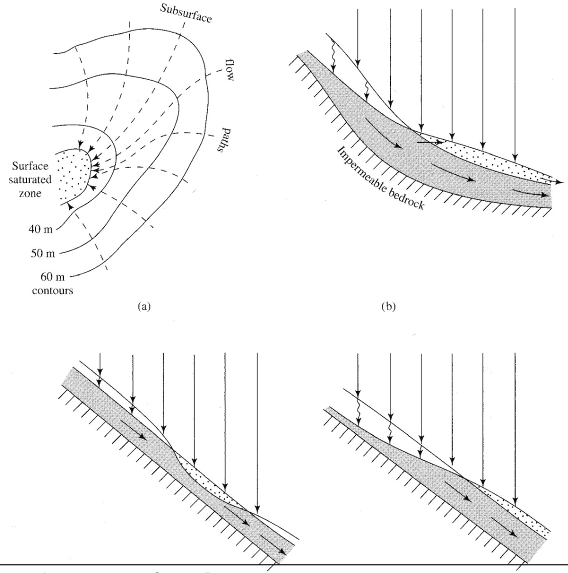
Variable Source Area (VSA)

A concept that saturation excess is the dominant contributor to runoff and extent of areas saturated from below varies widely, dependent on antecedent wetness.



Issues?

Mechanisms for Saturation Excess



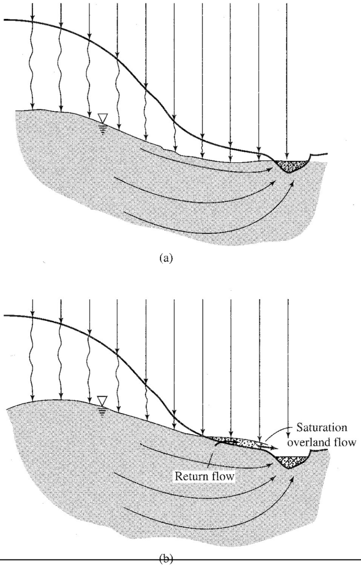
(a) (b) (c) (d)

- a. convergence
- b. slope break
- c. local thinning out of soil
- d. saturation excess

Subsurface Event (Storm) Flow

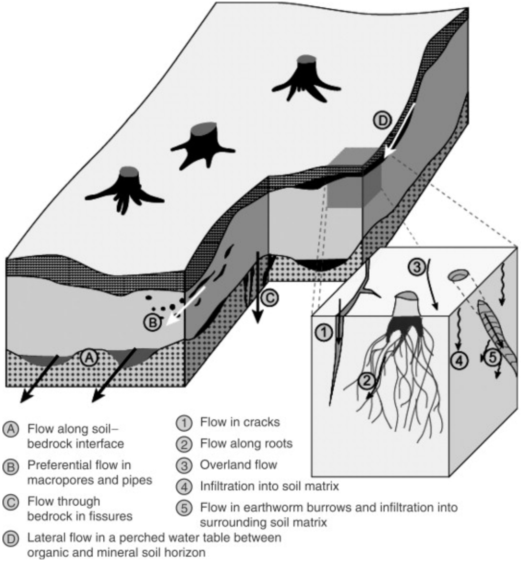
- Subsurface contribution to the response hydrograph that is hydraulically distinct from baseflow.
- Often referred to as **interflow**
- Can occur in the **saturated** and/or **unsaturated** zone.
- Likely the cause for the "Old Water Paradox."

Groundwater Mounding

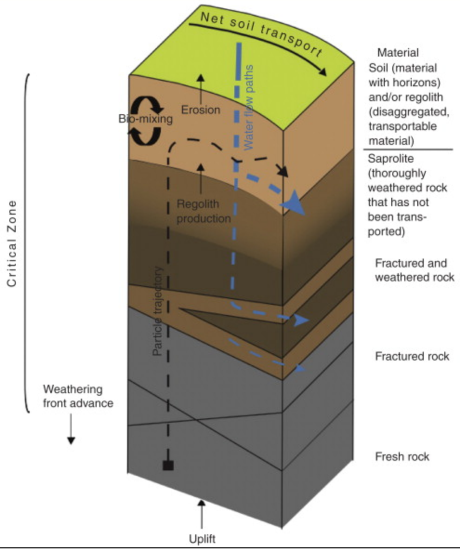


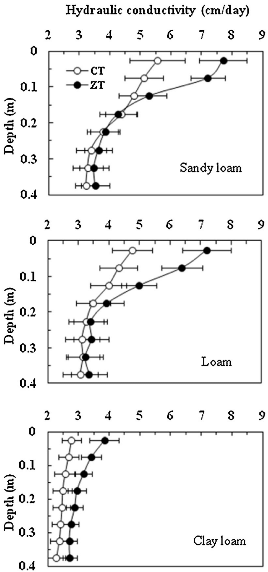
The same process that leads to saturation excess overland flow (infiltration of event water that leads to a rise in the saturated zone) also leads to larger subsurface contributions to streamflow due to steeper hydraulic gradients.

Perched Saturated Zones

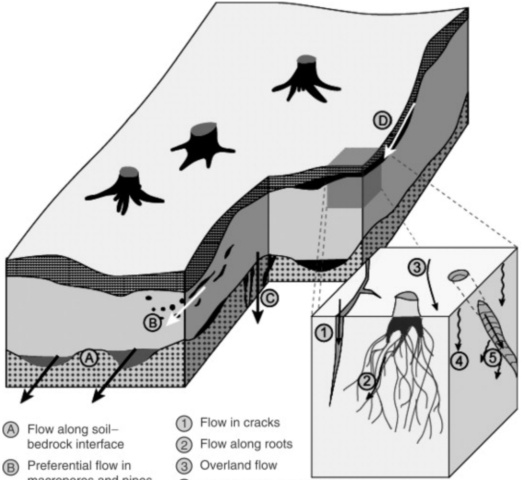


Hydraulic Conductivity vs Depth





Macropore Flow



- (A) Flow along soil–bedrock interface
- (B) Preferential flow in macropores and pipes
- (C) Flow through bedrock in fissures
- (D) Lateral flow in a perched water table between organic and mineral soil horizon
- (1) Flow in cracks
- (2) Flow along roots
- (3) Overland flow
- (4) Infiltration into soil matrix
- (5) Flow in earthworm burrows and infiltration into surrounding soil matrix